

Histopathologic changes in surficial tissues of brook trout, *Salvelinus fontinalis* (Mitchill), exposed to acute and chronic levels of pH

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Fingerling brook trout, *Salvelinus fontinalis* (Mitchill), were subjected to acute and chronic pH stress for maximum periods of 10 000 min in a series of continuous-flow dilutions of sulphuric acid or sodium hydroxide from pH 2.2 to 10.8 at 10 and 20 °C after acclimation at 15 °C and pH 6.8. Various tissues, gill, eye, naris, integument, and alimentary tract, were examined histologically and compared with control samples kept at pH 6.8. No differences in the degree or form of tissue injury were detected between series for corresponding pH levels at 10 and 20 °C. Thresholds for tissue and cellular derangements were pH 5.2 and 9.0. Mucous cells of gills, nares, and integument exhibited progressive degrees of hypertrophy and excessive secretion of mucus with increased pH stress. Epithelial necrosis and sloughing occurred extensively on gills, corneae, and integument. At the lethal levels (pH 3.5 and 9.8), epithelial necrosis also occurred in the esophagus. No cellular injury was detected in the stomach or any part of the intestine.

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Du fretilin de *Salvelinus fontinalis* (Mitchill) acclimaté à une température de 15 °C et à un pH de 6.8 a été soumis à des stress aigus et chroniques de pH, pour des périodes allant jusqu'à 10 000 min, dans une série de dilutions à écoulement continu d'acide sulfurique ou de soude caustique, à pH variant entre 2.2 et 10.8, à 10 et 20 °C. Plusieurs tissus, branchie, oeil, narine, tégument et canal alimentaire, ont subi un examen histologique et ont été comparés à des échantillons témoins provenant de poissons gardés à un pH de 6.8. Il n'y a pas de différence de gravité ou de forme des blessures tissulaires entre les séries à 10 et celles à 20 °C, à des pH correspondants. Les seuils de dommage cellulaire et tissulaire se situent à des pH de 5.2 et de 9.0. Les cellules muqueuses des branchies, des narines et du tégument réagissent à une augmentation du stress par une hypertrophie de plus en plus grande et une sécrétion excessive de mucus. On constate une nécrose épithéliale et une desquamation importantes des branchies, de la cornée et du tégument. Aux pH létaux (3.5 et 9.8), la nécrose épithéliale affecte aussi l'oesophage. On ne constate aucune blessure cellulaire à l'estomac ou à l'intestin.

[Traduit par le journal]

Introduction

Recent literature suggests that industrialization and urbanization in Europe and North America are modifying the pH of natural waters by air- and water-borne effluents, in some instances with concomitant changes in the ichthyofauna (EIFAC 1969; Jensen and Snekvik 1972; Almer 1972; Brossett 1973; Almer *et al.* 1974; Hogström 1973, 1974; Bormann 1975; Haapala *et al.* 1975; Li and Landsberg 1975). In Canada, Beamish and Harvey (1972), Beamish (1974), and Beamish *et al.* (1975) have shown increasing acidity in Ontario lakes and on the basis of fish mortality, they have classified lakes of pH 4.5-5.5 as endangered, and those below pH 4.5 as critically acidic. Daye and Garside (1975) reported the approximate lower and upper lethal

levels for the brook trout, *Salvelinus fontinalis* (Mitchill), to be pH 3.5 and 9.8 for maximum exposures of 10 000 min at 10 and 20 °C.

The nature of the fatal injury and the extent of sublethal injuries in fishes subjected to abnormal levels of pH have not been reported in detail. It is, therefore, the purpose of the present report to illustrate histologically these effects in the surficial tissues of brook trout reclaimed from the experiments of Daye and Garside (1975).

Materials and Methods

Hatchery-reared fingerling brook trout, *Salvelinus fontinalis* (Mitchill), were obtained from the Coldbrook Fish Culture Station, Coldbrook, Kings Co., Nova Scotia, for the study. Procedures for acclimation, experimental exposures, and determination of lethal levels of pH have been described by Daye and Garside (1975) for the fish used in this examination.

TABLE 1. Approximate thresholds of pH for injury to surficial tissues of fingerling brook trout, *Salvelinus fontinalis*, exposed for maximums of 10 000 min to several levels of pH at 10 and 20 °C

Tissue	pH											
	2.2	3.2	3.6	4.2	4.8	5.2	5.6	6.8	9.0	9.5	10.0	10.8
Gill	×	×	×	×	×	×	—	—	×	×	×	×
Integument (operculum)	×	×	×	×	—	—	—	—	×	×	×	×
Eye (cornea)	×	×	×	×	—	—	—	—	—	×	×	×
Naris	×	×	×	×	—	—	—	—	—	—	×	×
Integument (body)	×	×	—	—	—	—	—	—	—	×	×	×
Eye (lens)	×	—	—	—	—	—	—	—	—	×	×	×
Esophagus	×	—	—	—	—	—	—	—	—	—	—	×
Stomach	—	—	—	—	—	—	—	—	—	—	—	—
Intestine	—	—	—	—	—	—	—	—	—	—	—	—

NOTE: × injury; — no injury.

The moment of death of each fish was observed and the elapsed time noted. The criterion of death was the cessation of opercular movement. Surviving fish were killed by severing the spinal cord immediately posterior to the brain. Preparation of each specimen for histological examination was begun within 5 min of death. The viscera were fixed immediately by intraperitoneal injection of 10% neutral formalin, before total immersion for 1 h (Humason 1972). They were then dissected for recruitment of the gill, eye, naris, operculum, lateral integument, esophagus, stomach, and intestine. These tissues were immediately placed into fresh fixative for 24 h. The fixed specimens were dehydrated, cleared with xylene, impregnated with molten paraffin under a vacuum of 1.05 kg/cm² for 2 h, and embedded in fresh paraffin under a vacuum of 1.67 kg/cm² for 2 h. The embedded material was then placed in an oven at 37 °C for 24 h. Sections were cut at 5 µm, mounted on slides, and stained with hematoxylin and eosin. Permount and coverslips preserved the sections.

Results

Alterations of surficial tissues occurred at all levels of pH from 5.2 and lower, and 9.0 and higher. No differences in the severity of injuries were detected by visual appraisal between those exposed at 10 and at 20 °C. However, injuries became progressively more severe within tissues and progressively more tissue sites were injured, with increasing acidity and increasing alkalinity beyond the thresholds of 5.2 and 9.0 (Table 1). Those levels of pH that were lethal during 10 000 min are designated as 'acute' and those that produced obvious tissue injury but were not lethal in the allotted time are designated as 'chronic.'

The gills exhibited two types of reactions to pH stress. Hypertrophy of mucous cells was observed first at the base of the gill filaments with increased ion concentrations. Accumulations of mucus formed on the intact gills of surviving fish at these concentrations.

The major and most deleterious effect caused by such levels of pH occurred to the gill lamellae near the tips of filaments (Fig. 1). All lamellae were affected but there was a gradual decrease in the severity of injury with the proximity to the gill arch. The lamellae were injured at pH 9.0 and above, and below pH 5.6. The form of injury was identical for both acidic and alkaline conditions but for acidic conditions the degree of injury was greater. The endothelial and epithelial cells of the lamellae progressively separated from the supporting pilaster cells with increased stress. At extremes, the pilaster cells were exposed, leaving the endothelial and epithelial cells clumped and separated from the lamellae and filaments.

Injury to the cornea and lens of the eye occurred beyond the range of pH 4.8–9.0 (Figs. 2, 3). Extreme alkaline conditions produced more pronounced injury than extreme acidic conditions. Apparent blending of the squamous epithelial cells of the cornea occurred at pH 9.5 and 4.2. At pH 10.0 and 3.2, only minor fragments of the epithelial tissues remained. All cells above the basal level of the epithelium were destroyed with the remaining underlying cells being condensed and distorted. The cornea was devested of epithelial cells at pH 10.8. The substantia propria became thicker in acute alkaline conditions. It was not affected in extreme acidic conditions. The lens was affected in extreme alkaline water and to a minor degree in very acute acidity. The lens capsule and epithelium were affected at and above pH 9.5. The lens capsule lost its structural clarity. The epithelium developed a deranged cellular structure with a consequent blending of lens fibers, epithelium,

and capsule. An enlarged lens capsule at pH 2.2 was the sole acidic effect. No injuries were apparent in the ciliary body, iris, retina, and other components of the eye, at this level of observation.

The naris was unaffected within the range of pH 4.8–9.5. Extreme acidic conditions produced a more pronounced effect than extreme alkaline conditions (Fig. 4). Tissue injury may be generalized for all conditions by noting the sequence in which injury occurred in the oval folds of olfactory epithelium. The broader end of the oval structure exhibited the first cellular alteration, an increase in goblet cells. The narrow end of the oval structure had a smaller increase in the number of goblet cells. The long surfaces of the oval structure did not produce goblet cells in any of the tests.

Goblet cells of increased size and number were produced at the broader end of the epithelial folds within the olfactory rosette at pH 4.2. These goblet cells increased in number at lower levels of pH. Cells of the lamina propria developed a higher degree of vacuolization at pH 4.2. Intense vacuolization at pH 3.2 created cytoplasmic disruption resulting in the darker appearance of nuclear bodies. Extreme alkaline conditions, above pH 9.5, created changes in the naris similar to that created by acid stress, but in lesser degrees. Goblet cell size and number were both less in those disturbed by extreme acidity. The vacuolization of lamina propria was noted only for pH 10.8. Dark nuclear bodies were observed in small quantities at pH 10.0.

The integument was examined at two sites on the fish. The area designated 'body integument' included the region ventral to the dorsal fin and immediately dorsal to the lateral line. The second designated area included the external integument of the mid opercular region, which will be termed 'opercular integument,' and integument of the caudal margin of the operculum, 'opercular tip.'

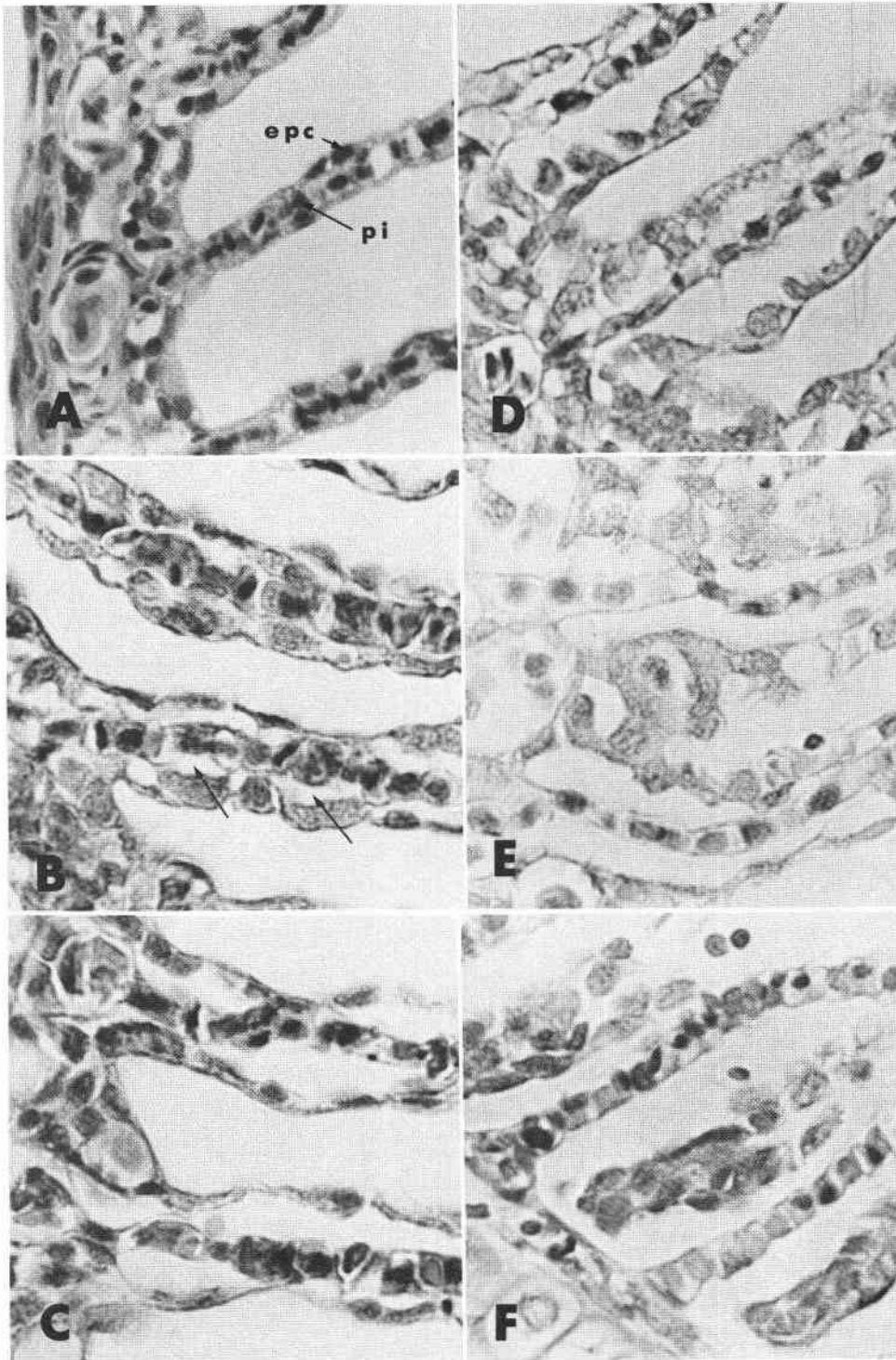
Body integument exhibited structural injury only in acutely high and low pH (Fig. 5). Glandular mucous cells became enlarged and distorted. Massive mucous secretions were evident in the preparations. Extreme alkaline conditions resulted in complete removal of the epithelium and damage to the exposed corium. Extreme acidic conditions caused corrosion of the external layers of stratified epithelium and distortion of the remaining epithelial cells. Chronic levels of pH did not injure the body integument. The opercular integument was damaged by chronic levels of pH in both extremes (Fig. 6). Cellular destruction was similar to that of the body integument, but the thresholds of pH stress were lower. The squamous epithelial layer was sloughed at pH 9.0 and at 3.6. Hypertrophy and gross distortion of mucous cells were especially pronounced in extreme acidic pH. In an extreme alkaline state, pH 10.8, the corrosive action of the alkaline water completely removed the epidermis and exposed the corium.

The integument of the opercular tip received greater injury than integument of the lateral operculum or that of the body (Fig. 7). At pH 4.2, the epithelial cells were disrupted. Epithelial damage increased with increased acidity as evidenced by connective tissue that appeared to be gelatinous with freely dispersed nuclei. Alkaline pH also caused damage to the connective tissue from pH 9.5. The glandular mucous cells also exuded copious mucus at this level. At pH 10.8, the epithelium was completely destroyed and exposed an injured corium. Alkaline alteration created a fibrous structure in the connective tissue, which appeared to be void of nuclei.

The esophagus was only affected under extremely acute pH stress (Fig. 8). The mucosa of the esophagus exhibited dark condensed nuclear bodies and cellular vacuolization at pH 2.2 and 10.8.

Examination of histological preparations revealed that the stomach and the entire length of

FIG. 1. Gill lamellae. Photomicrographs of longitudinal sections of gills from the brook trout, *Salvelinus fontinalis*, illustrating the junctions of the gill lamellae and the gill filament. *pi*, pilaster cell; *epc*, epithelial cell. (A) pH 6.8. Control. Normal cellular structure of gill at junction between lamellae and filament. $\times 750$. (B) pH 9.0. Arrows indicate areas of separation between epithelial cells and pilaster cells. $\times 750$. (C) pH 9.5. Higher degree of separation between epithelial cells and pilaster cells. $\times 750$. (D) pH 5.2. Epithelial cells partly separated from pilaster cells. $\times 750$. (E) pH 3.2. Separation of epithelial cells from pilaster cells virtually complete. Cells at junction are attached. $\times 750$. (F) pH 2.2. Epithelial cells completely separated from pilaster cells and appear in a clumped 'v' formation. $\times 750$.



the intestine were unaffected at all levels of experimental pH that were used in this study.

Discussion

After publication of data on tolerance at low pH in brook trout (Daye and Garside 1975), in which sulphuric acid was used as the source of hydrogen ions, colleagues raised the question of the possible extent to which morbidity was induced by sulphate ions rather than the hydrogen ions. This objection was sufficiently met by subjecting samples of fingerling brook trout for 10 000 min to solutions of sodium, magnesium, and calcium sulphate, adjusted to yield the same concentrations of sulphate ions as those in the original tests in which sulphuric acid was used (maximum concentration equivalent to 28.5 mg/l). There was no mortality during any of these subsequent trials and no outward indications of distress or production of moribund superficial tissue in any of the subjects. Several diverse fishes, generally considered to be freshwater species, survive sulphate levels more than 20

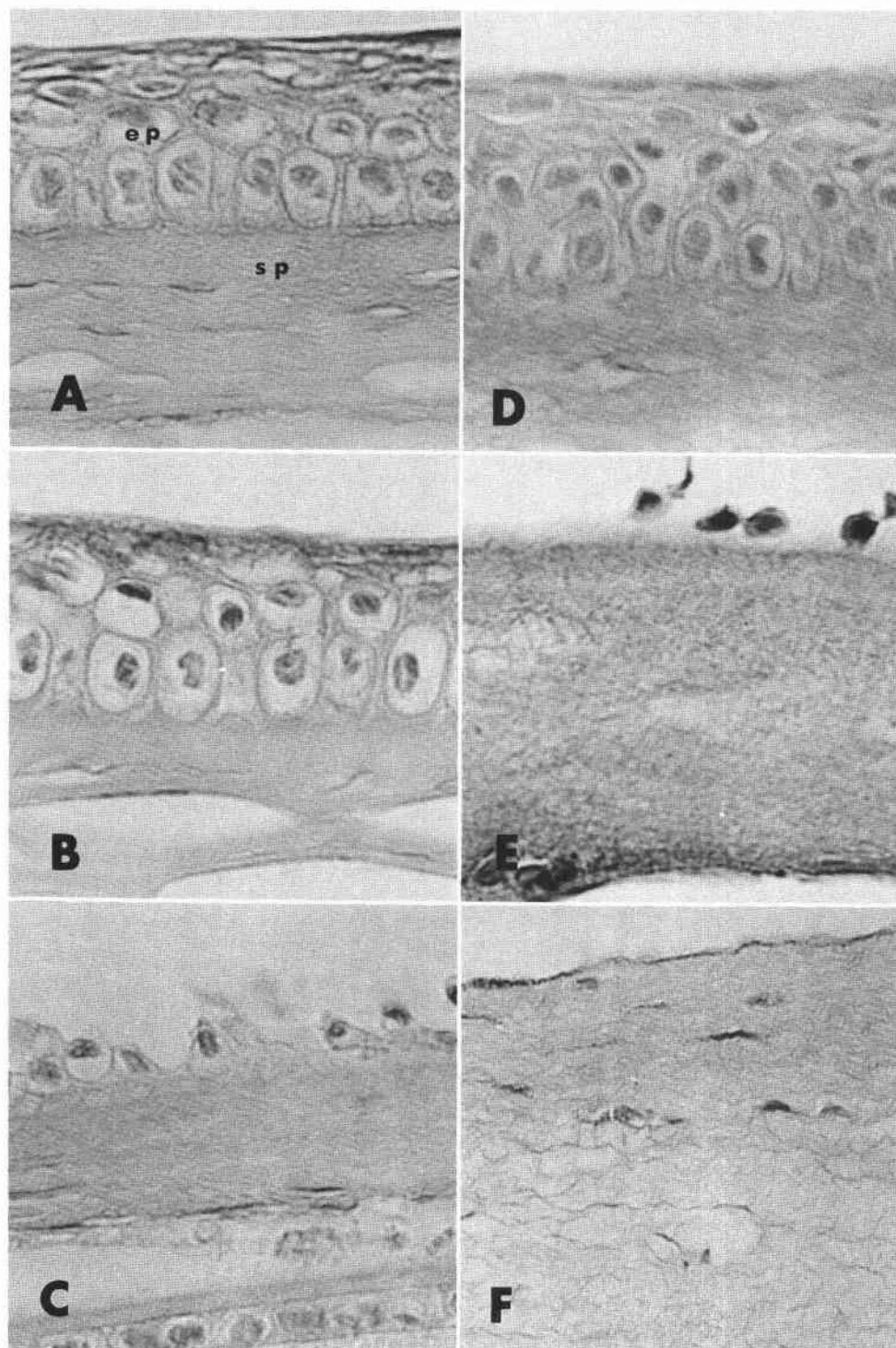
times the concentrations described in our study (Rawson and Moore 1944). Also, McKee and Wolf (1963) reported levels of sulphate ions exceeding 32 mg/l in 50% of inland waters of the United States that support 'good' game-fish populations. Therefore, we have concluded that our observations of structural alterations in this report are unrelated to the presence of sulphate ions.

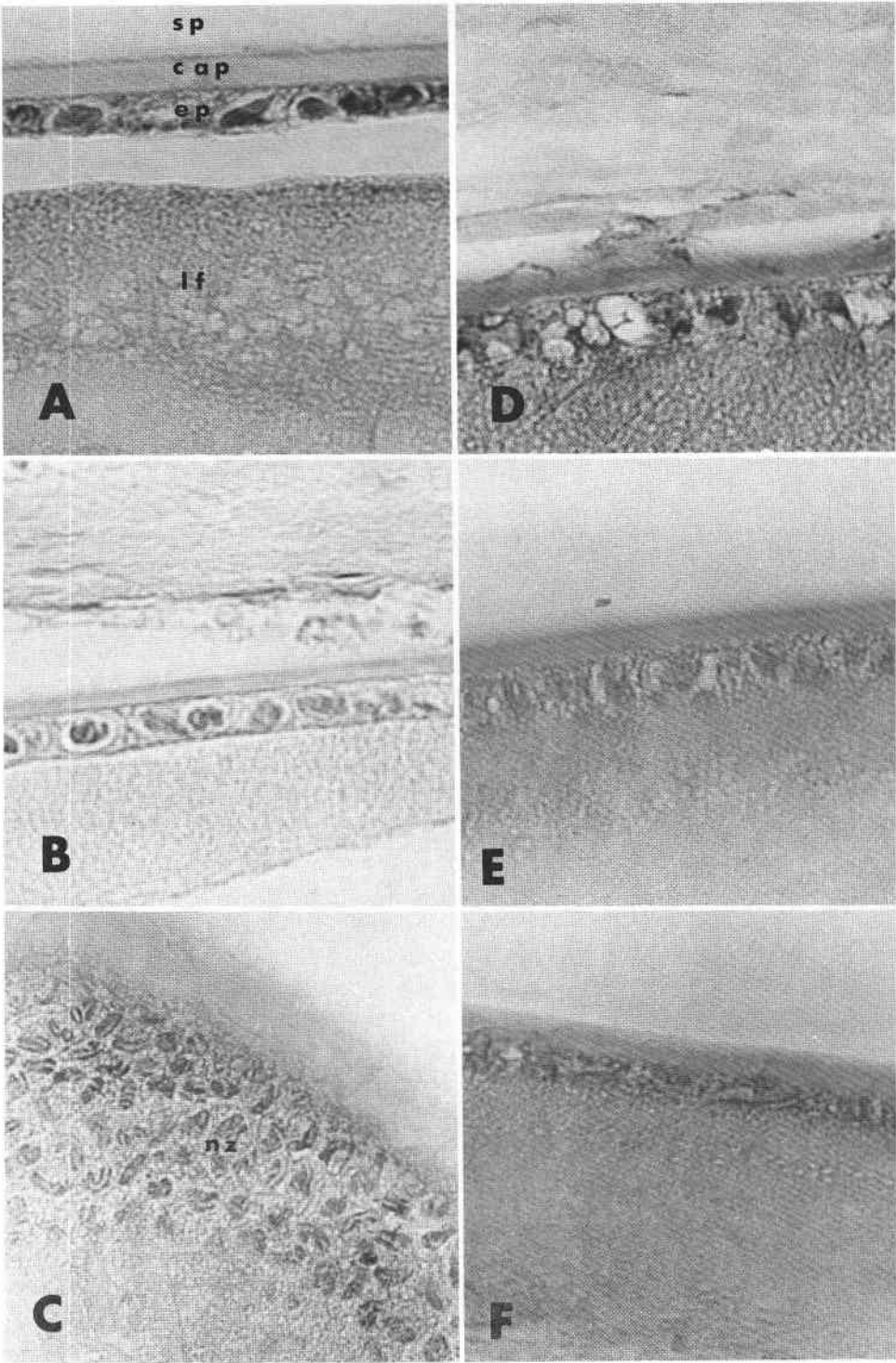
Excessive mucous secretions have been reported by Ellis (1937), Westfall (1945), Plonka and Neff (1969), Vaala *et al.* (1969), Vaala and Mitchell (1970), and Anthony *et al.* (1971) for fishes under pH stress. These reports are supported in the present study by observed hypertrophy of mucous cells on gill filaments and the consequential secretion of mucus. Excessive secretions have also been noted for fishes subjected to heavy metals (Dilling *et al.* 1926; Carpenter 1927; Ellis 1937, 1947; Jones 1938, 1939; Schweiger 1957; Ashley 1970; Eisler and Gardner 1973; and Eisler 1974), and to pesticides (Christie and Battle 1963).

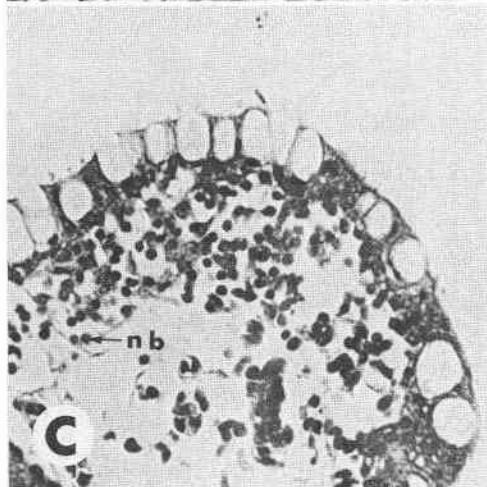
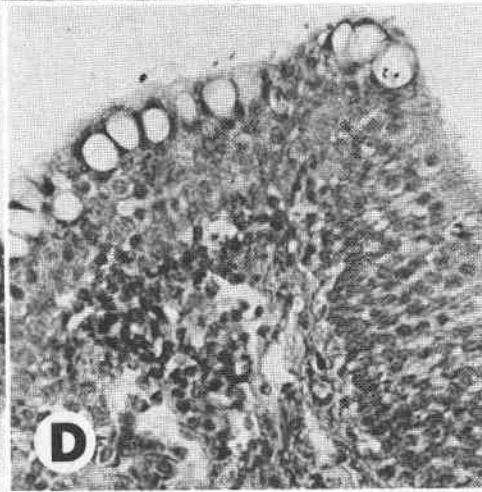
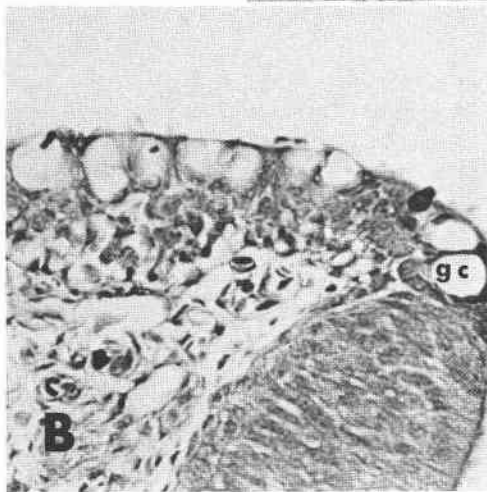
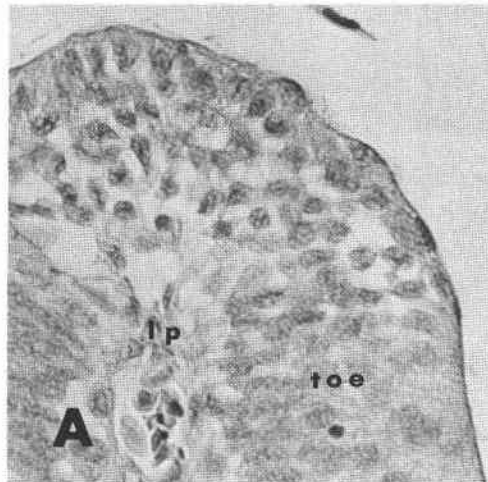
FIG. 2. Eye cornea. Photomicrographs of cornea cross sections from the brook trout, *Salvelinus fontinalis*. *ep*, epithelium; *sp*, substantia propria. (A) pH 6.8. Control. Normal cornea with a distinct division between epithelium and substantia propria. $\times 750$. (B) pH 4.2. Outer layers of epithelial tissue appear blended. $\times 750$. (C) pH 3.2. Acute damage to corneal epithelium. Most of the epithelial cells are lost. Remaining cells are irregular in shape. $\times 750$. (D) pH 9.5. Region between substantia propria and epithelium not distinct. $\times 750$. (E) pH 10.0. Most of the epithelial tissue lost. Remaining cells are condensed in size and are irregular in shape. $\times 750$. (F) pH 10.8. Epithelial tissue completely lost. Substantia propria thickened considerably. $\times 500$.

FIG. 3. Eye lens. Photomicrographs of lens capsule and epithelial cross sections from the brook trout, *Salvelinus fontinalis*. *ep*, epithelium; *sp*, substantia propria; *cap*, capsule; *lf*, lens fibres; *nz*, nuclear zone. (A) pH 6.8. Control. Section directly medial to corneal center. Normal distinct lens capsule and single layer of epithelium. Separation of fibers from epithelium is due to handling of fragile tissue. $\times 750$. (B) pH 3.2. Section directly medial to corneal center. All components of lens normal and distinct. $\times 750$. (C) pH 2.2. Section medial to the cornea and near the tip of the iris. Capsule of lens thickened and not distinct. Epithelium lost. Nuclear zone pronounced. $\times 500$. (D) pH 9.5. Section directly medial to corneal center. Indistinct structure of lens epithelium. Cellular appearance abnormal. Lens capsule and fibers are normal. $\times 750$. (E) pH 10.0. Section directly medial to corneal center. Lens capsule indistinct. Epithelial cells grossly distorted. $\times 750$. (F) pH 10.8. Section directly medial to corneal center. Lens capsule and epithelial layer of cells are indiscernible. Lens fibers are normal. $\times 750$.

FIG. 4. Naris. Photomicrographs of naris cross sections from the brook trout, *Salvelinus fontinalis*. The periphery of a single fold of olfactory epithelium within the olfactory rosette. *gc*, goblet cell; *lp*, lamina propria; *toe*, true olfactory epithelium; *nb*, nuclear body. (A) pH 6.8. Control. Segment of the broader end of an oval fold of olfactory epithelium. Normal tissue. Lack of prominent goblet cells. Lamina propria not highly vacuolized. $\times 500$. (B) pH 4.2. Segment of the broader end of an oval fold of olfactory epithelium. Hypertrophy of goblet cells. Increased vacuolization of the lamina propria. $\times 300$. (C) pH 3.2. Segment of the broader end of an oval fold of olfactory epithelium. Hypertrophy of goblet cells more pronounced. High degree of vacuolization within the lamina propria. Distinct, dark, condensed nuclear bodies. General cellular disarray of olfactory epithelium. $\times 300$. (D) pH 10.0. Segment of the broader end of an oval fold of olfactory epithelium. Hypertrophy of goblet cells. Lamina propria slightly vacuolized. $\times 300$. (E) pH 10.8. Segment of the broader end of an oval fold of olfactory epithelium. Slight hypertrophy of goblet cells. Lamina propria highly vacuolized. General cellular disarray of olfactory epithelium. $\times 300$.







Gill irritation was observed by Lewis and Peters (1956) in fish in waters of pH 4.0–4.5. Injury to gill epithelium is a common response to extreme levels of environmental pollutants and other variables as discussed by Bandt (1936) for high pH levels, Lloyd (1960), Malacea (1966), Baker (1969), Gardner and Yevich (1970), and Skidmore (1970) for heavy metals, Odense *et al.* (1972) for phosphorus, Gardner *et al.* (1975) for light oil, and Schmid and Mann (1961) for detergents. Apparently the present study provides the first description of necrosis and desquamation of epithelium from gill lamellae that subsequently exposes the supporting pillar cells and capillaries to the ambient medium, as a result of exposure to pH stress. Similar epithelial sloughing of the gill lamellae has been reported by Schweiger (1957), Lloyd (1965), and Skidmore and Tovell (1972) for heavy metals, and by Mitrovic *et al.* (1968) for phenols. This form of gill injury is rather different than swollen lamellae, a condi-

tion frequently reported for detergents (Lemke and Mount 1963; Brown *et al.* 1968), organic herbicides, and pesticides (Christie and Battle 1963; Cope 1965; McCraren *et al.* 1969), "gas bubble" disease (Pauley and Nakatani 1967), and many bacterial and nutritional diseases (Wood 1960).

Only two reports have been located that describe the occurrence of eye injury in fish exposed to extreme levels of pH. Eicher (1946) concluded that instances of widespread blindness in fish inhabiting some Arizona lakes must be due to high pH. In rainbow trout, *Salmo gairdneri*, Carter (1964) reported exophthalmic blindness below pH 5.0 in seawater to which sulphuric or hydrochloric acid was added. The present study is apparently the first to document the histologic nature of injury to the lens in addition to that of the cornea, an alteration which might well have been anticipated.

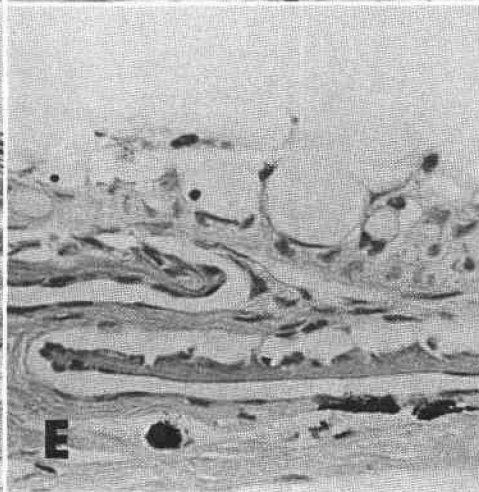
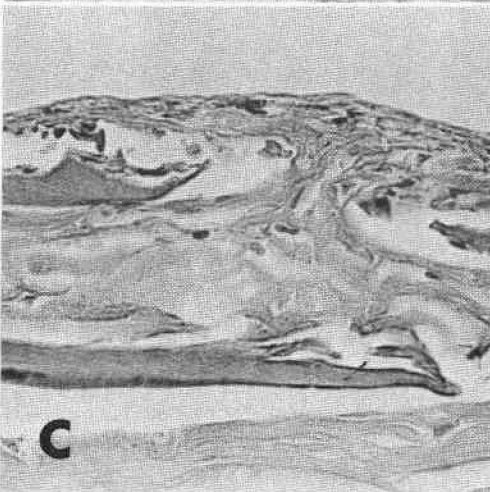
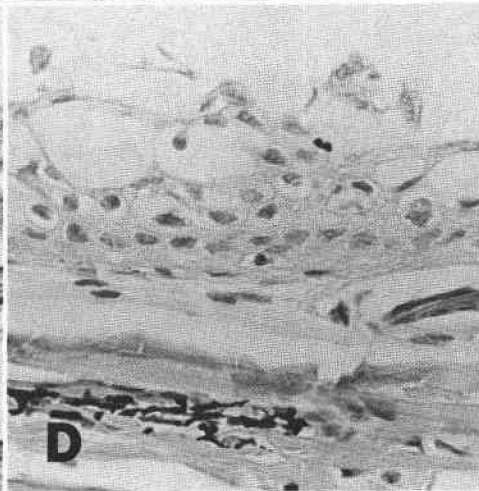
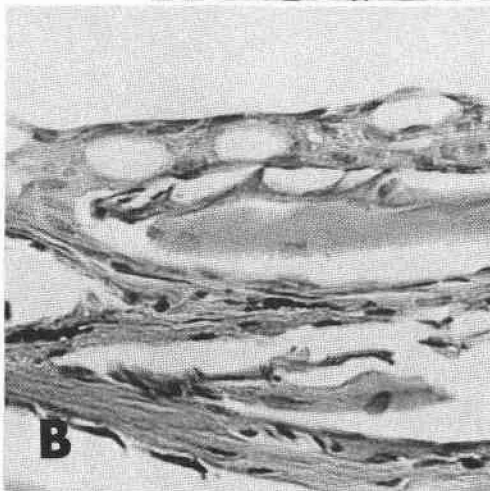
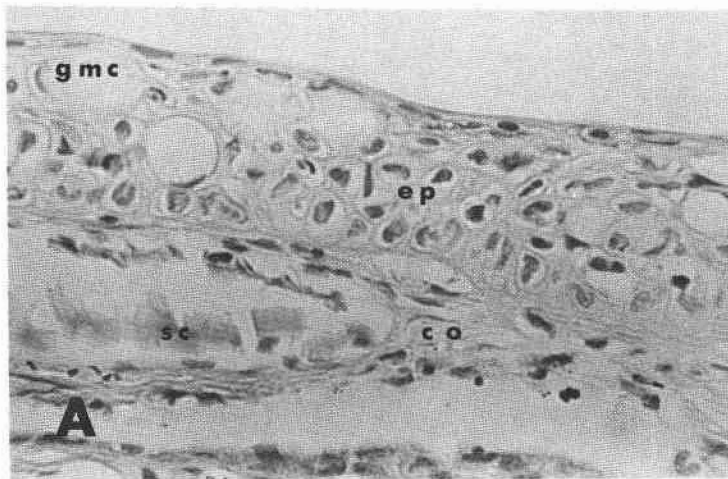
The compensatory mechanisms of the naris

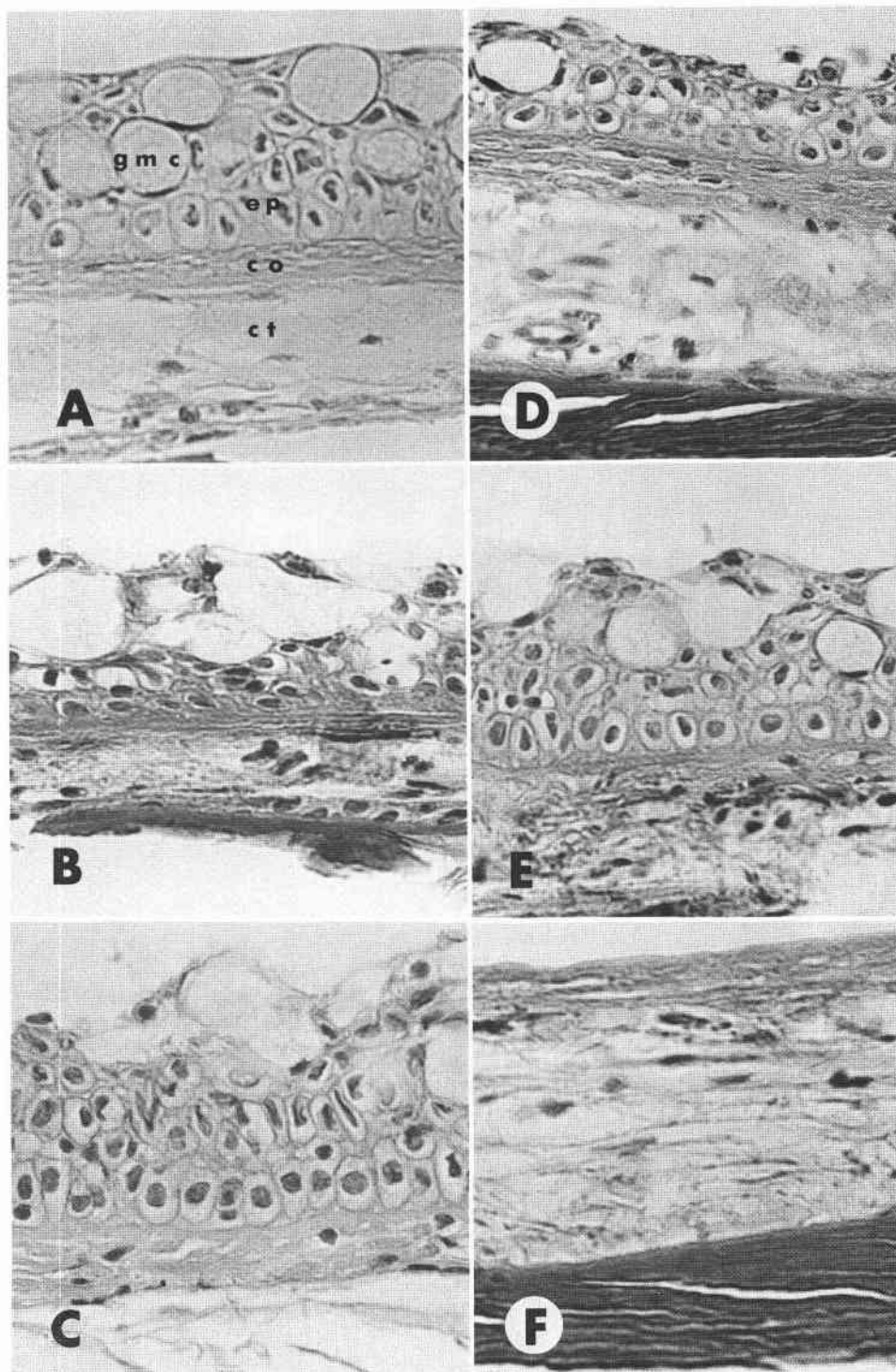
FIG. 5. Body integument. Photomicrographs of body integument cross sections from the brook trout, *Salvelinus fontinalis*. Tissues taken from the region ventral to the dorsal fin and immediately dorsal to the lateral line. *ep*, epithelium; *sc*, scale; *co*, corium; *gmc*, glandular mucous cell. (A) pH 6.8. Control. Normal body integument. Epithelium with distinct cellular structure. $\times 500$. (B) pH 10.0. Epithelium reduced in thickness. Glandular mucous cells are intact and have normal size. Remaining epithelial cells are indistinct from each other with a loss of cellular clarity. $\times 500$. (C) pH 10.8. Epithelium lost. Corium lacks cellular definition. Scales remain intact. $\times 500$. (D) pH 3.2. External layer of epithelial cells lost. Structural distortion of remaining epithelial cells. Glandular mucous cells enlarged with contents extruded. $\times 500$. (E) pH 2.2. External layer of epithelial cells lost. Severe structural distortion of remaining epithelial cells. Glandular mucous cells enlarged with many having contents extruded. $\times 500$.

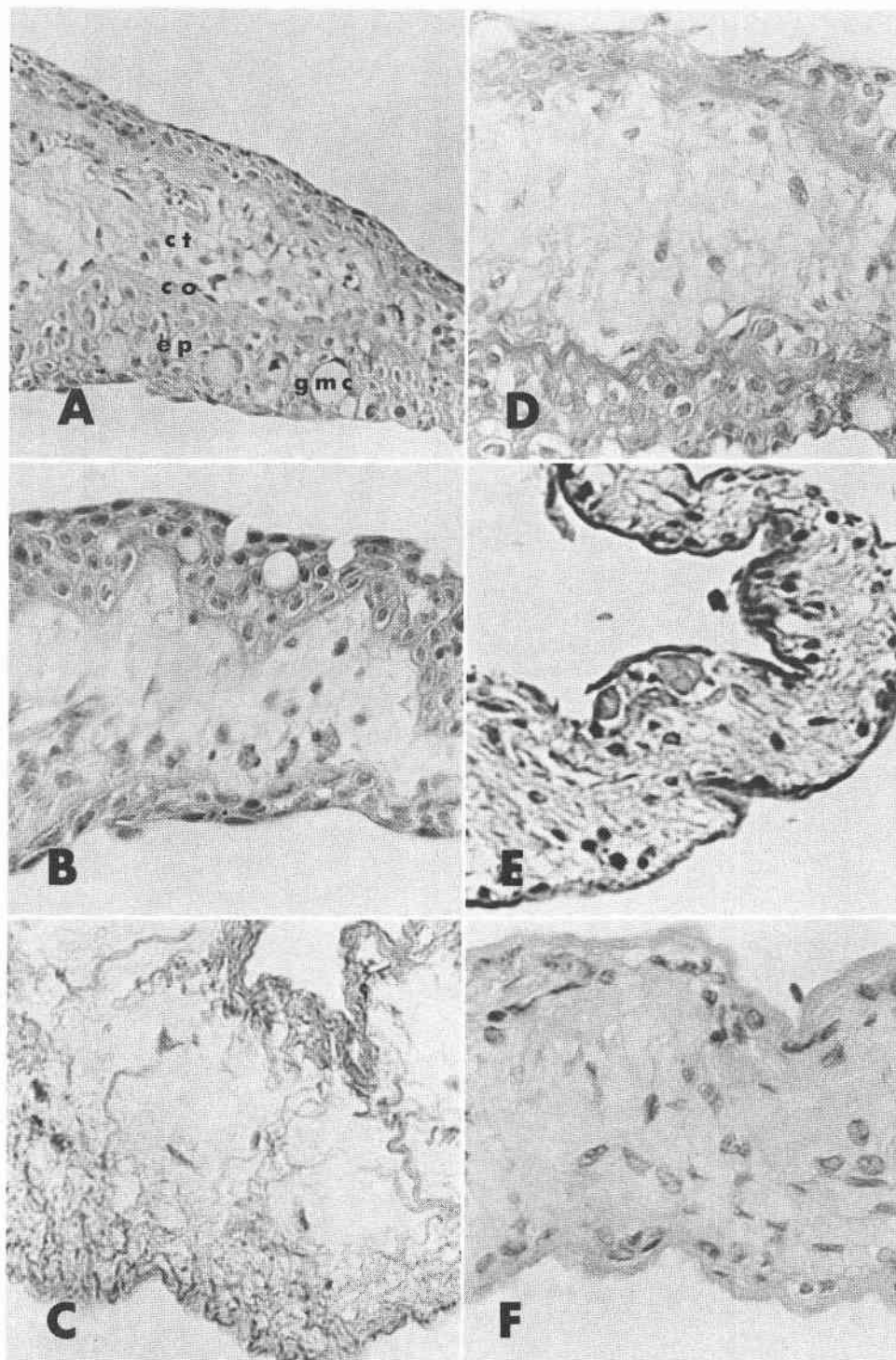
FIG. 6. Opercular integument. Photomicrographs of opercular integument cross sections from the brook trout, *Salvelinus fontinalis*. Integument of the midopercular region is illustrated. *ep*, epithelium; *co*, corium; *gmc*, glandular mucous cell; *ct*, connective tissue. (A) pH 6.8. Control. Opercular integument normal. $\times 500$. (B) pH 3.2. External layers of epithelium lost. Gross distortion and enlargement of glandular mucous cells. $\times 500$. (C) pH 2.2. External layers of epithelium lost. Glandular mucous cells exhibit increase in size, gross distortion, and extrusion of mucus. $\times 500$. (D) pH 9.0. External layer of epithelial cells disrupted. $\times 500$. (E) pH 9.5. External layer of epithelial cells disrupted. Glandular mucous cells in process of mucus extrusion. $\times 500$. (F) pH 10.8. Epithelium totally lost. Corium exposed. $\times 500$.

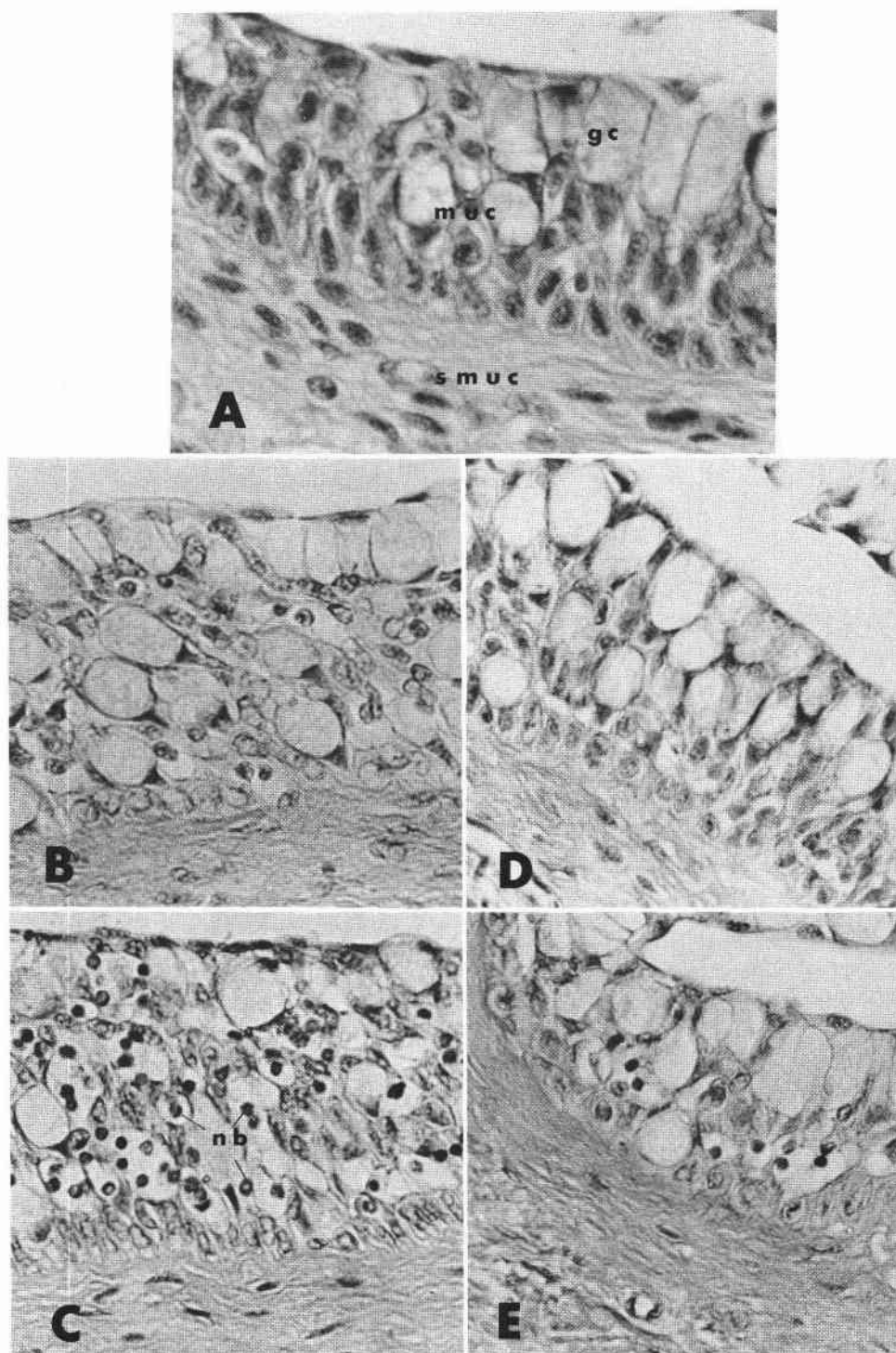
FIG. 7. Integument of the opercular tip. Photomicrographs of operculum cross sections from the brook trout, *Salvelinus fontinalis*. Integument of the most caudal region of the opercular tip is illustrated. *ep*, epithelium; *co*, corium; *gmc*, glandular mucous cell; *ct*, connective tissue. (A) pH 6.8. Control. Normal integument. $\times 300$. (B) pH 9.5. Connective tissue with structural deterioration. Exposed glandular mucous cell extruding mucus. $\times 500$. (C) pH 10.8. Epithelium destroyed. Exposed corium extensively damaged. Connective tissue with fibrous structure and free of nuclei. $\times 500$. (D) pH 4.2. External layer of epithelial cells disrupted. Connective tissue normal. $\times 500$. (E) pH 3.2. Epithelium lost. Exposed corium reduced. Connective tissue lacking structural definition. $\times 600$. (F) pH 2.2. Epithelium lost. Corium exposed. Connective tissue appears gelatinous with freely dispersed nuclei. $\times 500$.

FIG. 8. Esophagus. Photomicrographs of esophagus cross sections from the brook trout, *Salvelinus fontinalis*. Anterior esophageal region illustrating the mucosa. *gc*, goblet cell; *muc*, mucosa; *smuc*, submucosa; *nb*, nuclear body. (A) pH 6.8. Normal stratified low columnar epithelium of the mucosa. $\times 500$. (B) pH 3.2. Occasional appearance of nuclear bodies. $\times 500$. (C) pH 2.2. Distinct dark nuclear bodies prominent. Vacuoles enlarged. $\times 500$. (D) pH 9.5. $\times 500$. (E) pH 10.8. Distinct dark condensed nuclear bodies prominent. Vacuoles enlarged. Cellular organization of mucosa lacking. $\times 500$.









that react against extreme levels of pH are different from those that react against heavy metals. Gardner and LaRoche (1973) and Gardner *et al.* (1974) in their studies of marine and estuarine fishes noted lesions of the olfactory surfaces exposed to heavy metals. These lesions differed somewhat according to the specific metal and none produced the extreme hypertrophy and vacuolization that characterizes exposure to low and high levels of pH.

Bandt (1946) was apparently the first to suggest that acid and alkaline waters could kill fish by a "corroding action" on the body surfaces and gills. Indeed, such an action has been demonstrated in the present study. Within the time allowance in these tests, only acute levels caused extensive necrosis and sloughing of the stratified epiderm and underlying corium. Protective responses by mucous cells occurred at most chronic levels, perhaps offsetting some of the injurious effects of the active ions. In general, the integument was more resistant to chronic pH levels than the other surfaces examined. The obviously more delicate integumentary fold extending from the opercular margins was correspondingly more sensitive than the integument overlying the lateral musculature.

The observed hypertrophy of mucous cells in the body integument accounts for the massive mucous secretions over the fish body observed by Westfall (1945), Trama (1954), Plonka and Neff (1969), and Dolsky and Anthony (1973), in studies with low pH that agree with our results. Anthony *et al.* (1971) noted that in brook trout, the mucous film over the entire body occurred only at levels of less than pH 4.0. Similarly, Beamish (1972) noted that when a mucous film had appeared on the body surface the fish did not revive when the stress was removed, indicating that copious body mucus is a symptom of incipient lethality. The buccal epithelium was not examined histologically. The mucosa of the esophagus was the only further portion of the alimentary tract affected by pH stress. The apparent failure of extreme levels of pH used in this study to injure either the gastric or intestinal mucosa is predictable because of the buffering capacity of the normal digestive secretions.

Each pair of physiological pO_2 and pH values creates a defined membrane potential (Laurent 1969). A shift in either would alter the membrane permeability and seriously alter the respiratory exchanges of oxygen and carbon dioxide at the

gill, such as those described by Wiebe *et al.* (1934), Townsend and Cheyne (1944), and Burton *et al.* (1972). This would reduce the scope (as defined by Fry 1947) for metabolic processes and consequential activities. Visual impairment is second only in importance, among those structures examined in this study, to respiratory debilitation. Injury to the cornea and lens would reduce the capability of the fish to feed and to avoid predators. Olfaction is also an important if not vital avenue of information for extended existence of fish. The stimulation of excessive mucous production at most chronic levels must impair the olfactory function.

Brook trout survive in natural waters in eastern Canada that frequently exceed the acidic threshold for tissue injury, particularly that for gills. Perhaps, then, in states of less severe irritation and injury, an equilibrium becomes established between cell destruction and replacement so that survival, although more precarious, is possible. No such process could be identified histologically in the present study but the experimental interval may not have been sufficiently long to accommodate the initiation of regenerative processes.

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